

Name :- Mihir Mungara

Snowflake Assignment-8 :-

Design a Secure Snowflake Environment for a Hypothetical Company

1. Created Roles

```
-- Step 1: Create Roles
CREATE OR REPLACE ROLE ADMIN;
CREATE OR REPLACE ROLE ANALYST;
CREATE OR REPLACE ROLE VIEWER;

SHOW ROLES;
```

Results :-

Results (just now)											
Table Chart Pivot created_on DESC x											
	created_on	name	is_default	is_current	is_inherited	assigned_to_users	granted_to_roles	granted_roles	owner	comment	
1	2026-06-04 05:35:09.766 -0700	VIEWER	N	N	N	0	0	0	ACCOUNTADMIN		
2	2026-06-04 05:35:08.367 -0700	ANALYST	N	N	N	0	0	0	ACCOUNTADMIN		
3	2026-06-04 05:35:08.210 -0700	ADMIN	N	N	N	0	0	0	ACCOUNTADMIN		

2. Created Database Objects

```
-- Step 2: Create Database Objects
CREATE OR REPLACE DATABASE SALES_DB;

CREATE OR REPLACE SCHEMA SALES_DB.PUBLIC;

CREATE OR REPLACE WAREHOUSE COMPUTE_WH
WITH WAREHOUSE_SIZE = 'XSMALL'
AUTO_SUSPEND = 60
AUTO_RESUME = TRUE;
```

3. Created Sample Tables

```
-- Step 3: Create Sample Tables
USE DATABASE SALES_DB;
USE SCHEMA PUBLIC;

CREATE OR REPLACE TABLE EMPLOYEES (
  ID INT,
  NAME STRING,
  SSN STRING,
  SALARY NUMBER(10,2)
);

INSERT INTO EMPLOYEES VALUES
(1, 'Alice', '123-45-6789', 50000),
(2, 'Bob', '987-65-4321', 60000);

SELECT * FROM EMPLOYEES;
```

Results (just now)

TableChartPivot

2 rows218ms

# ID	NAME	SSN	SALARY
1	Alice	123-45-6789	50000.00
2	Bob	987-65-4321	60000.00

4. Granted Privileges to Roles

```
-- Step 4: Grant Privileges to Roles

-- Admin Role
GRANT ALL PRIVILEGES ON DATABASE SALES_DB TO ROLE ADMIN;
GRANT ALL PRIVILEGES ON ALL SCHEMAS IN DATABASE SALES_DB TO ROLE ADMIN;
GRANT ALL PRIVILEGES ON ALL TABLES IN DATABASE SALES_DB TO ROLE ADMIN;
GRANT USAGE, OPERATE ON WAREHOUSE COMPUTE_WH TO ROLE ADMIN;
GRANT CREATE USER ON ACCOUNT TO ROLE ADMIN;
GRANT CREATE ROLE ON ACCOUNT TO ROLE ADMIN;
GRANT CREATE WAREHOUSE ON ACCOUNT TO ROLE ADMIN;

-- Analyst Role
GRANT USAGE ON DATABASE SALES_DB TO ROLE ANALYST;
GRANT USAGE ON SCHEMA SALES_DB.PUBLIC TO ROLE ANALYST;
GRANT SELECT ON ALL TABLES IN SCHEMA SALES_DB.PUBLIC TO ROLE ANALYST;
GRANT USAGE ON WAREHOUSE COMPUTE_WH TO ROLE ANALYST;

-- Viewer Role
CREATE OR REPLACE VIEW DASHBOARD_VIEW AS
SELECT ID, NAME, SALARY
FROM EMPLOYEES;

GRANT USAGE ON DATABASE SALES_DB TO ROLE VIEWER;
GRANT USAGE ON SCHEMA SALES_DB.PUBLIC TO ROLE VIEWER;
GRANT SELECT ON VIEW DASHBOARD_VIEW TO ROLE VIEWER;|
GRANT USAGE ON WAREHOUSE COMPUTE_WH TO ROLE VIEWER;
```

5. Created User and Assigned Default Roles

```
-- Step 5: Create Users

-- Admin User
CREATE OR REPLACE USER ADMIN_USER
PASSWORD='Admin@123'
DEFAULT_ROLE=ADMIN
DEFAULT_WAREHOUSE=COMPUTE_WH
MUST_CHANGE_PASSWORD=TRUE;

-- Analyst User
CREATE OR REPLACE USER ANALYST_USER
PASSWORD='Analyst@123'
DEFAULT_ROLE=ANALYST
DEFAULT_WAREHOUSE=COMPUTE_WH
MUST_CHANGE_PASSWORD=TRUE;

-- Viewer User
CREATE OR REPLACE USER VIEWER_USER
PASSWORD='Viewer@123'
DEFAULT_ROLE=VIEWER
DEFAULT_WAREHOUSE=COMPUTE_WH
MUST_CHANGE_PASSWORD=TRUE;
```

6. Assigned Roles to Users

```
-- Step 6: Assign Roles to Users
GRANT ROLE ADMIN TO USER ADMIN_USER;
GRANT ROLE ANALYST TO USER ANALYST_USER;
GRANT ROLE VIEWER TO USER VIEWER_USER;

SHOW GRANTS TO USER ADMIN_USER;
SHOW GRANTS TO USER ANALYST_USER;
SHOW GRANTS TO USER VIEWER_USER;
```

7. Configured Network Policies

```
-- Step 7: Configure Network Policies
-- Unset existing network policy from account before replacing
ALTER ACCOUNT UNSET NETWORK_POLICY;

-- Create Network Policy
CREATE OR REPLACE NETWORK POLICY SECURE_POLICY
ALLOWED_IP_LIST = (
  '192.168.1.0/24',
  '203.0.113.0/32',
  '49.200.158.22'
)
BLOCKED_IP_LIST = (
  '10.0.0.0/8'
);

SHOW NETWORK POLICIES;

ALTER ACCOUNT SET NETWORK_POLICY = SECURE_POLICY;

SHOW PARAMETERS LIKE 'NETWORK_POLICY' IN ACCOUNT;
```

Results (just now)							
Table Chart Pivot							
	created_on	name	comment	# entries_in_allowed_ip_list	# entries_in_blocked_ip_list	# entries_in_allowed_network_rules	# entries_in_blocked_network_rules
1	2026-06-04 05:40:21.	SECURE_POLICY		3	1	0	0

8. MFA needs to be enabled by user end

9. Implemented Data Masking

```
-- Step 9: Implement Data Masking
-- Create Masking Policy
CREATE OR REPLACE MASKING POLICY SSN_MASK
AS (VAL STRING)
RETURNS STRING ->
CASE
    WHEN CURRENT_ROLE() = 'ADMIN'
    THEN VAL
    ELSE 'XXX-XX-XXXX'
END;

ALTER TABLE EMPLOYEES
MODIFY COLUMN SSN
SET MASKING POLICY SSN_MASK;

DESC TABLE EMPLOYEES;
```

10. Tested Data Masking

```
163 -- Step 10: Test Data Masking
164 -- Admin Access
165 USE ROLE ADMIN;
166 SELECT * FROM EMPLOYEES;
167
168 -- Analyst Access
169 USE ROLE ANALYST;
170 SELECT * FROM EMPLOYEES;
171
172 -- Viewer Access
173 USE ROLE VIEWER;
174 SELECT * FROM DASHBOARD_VIEW;
```

Results (just now)

#	ID	NAME	SSN	SALARY
1	1	Alice	123-45-6789	50000.00
2	2	Bob	987-65-4321	60000.00

```
163 -- Step 10: Test Data Masking
164 -- Admin Access
165 USE ROLE ADMIN;
166 SELECT * FROM EMPLOYEES;
167
168 -- Analyst Access
169 USE ROLE ANALYST;
170 SELECT * FROM EMPLOYEES;
171
172 -- Viewer Access
173 USE ROLE VIEWER;
174 SELECT * FROM DASHBOARD_VIEW;
```

Results (just now)

#	ID	NAME	SSN	SALARY
1	1	Alice	XXX-XX-XXXX	50000.00
2	2	Bob	XXX-XX-XXXX	60000.00

```
163 -- Step 10: Test Data Masking
164 -- Admin Access
165 USE ROLE ADMIN;
166 SELECT * FROM EMPLOYEES;
167
168 -- Analyst Access
169 USE ROLE ANALYST;
170 SELECT * FROM EMPLOYEES;
171
172 -- Viewer Access
173 USE ROLE VIEWER;
174 SELECT * FROM DASHBOARD_VIEW;
```

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Results (just now)

[Table](#) [Chart](#) [Pivot](#)

2 rows 138ms

# ID	NAME	SALARY
1	Alice	50000.00
2	Bob	60000.00